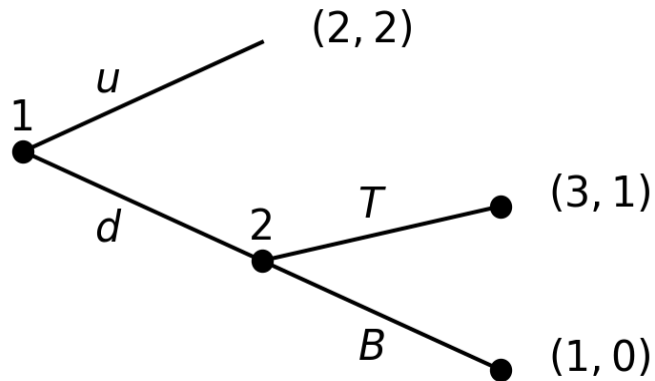


Equilibrium refinements in extensive-form games

Subgame perfection and non-credible threats



	T	B
u	2,2	[2,2]
d	[3,1]	1,0

-
- Two Nash equilibria
 - (d, T) .
 - (u, B) . Credible?
 - Subgame perfection eliminates noncredible threats in games of perfect information.

Subgame of a game

- Let x be a node in $(X, \succ)$, the game tree in an extensive form game Γ . The set of nodes $X' = \{y : y \succ x\}$ is a subgame, denoted by $\Gamma(x)$, if X' with the structure inherited from Γ , is an extensive form game.

Remarks

1. The information set containing x must be a singleton.
2. For every information set u in Γ , either $u \subset S$ or $u \cap X' = \emptyset$.

Let Γ be an extensive form game with I players.

- For any node x , a behavior-strategy profile $\sigma = (\sigma_1, \dots, \sigma_I)$ determines the probability $P_\sigma(x)$ that x will be reached.

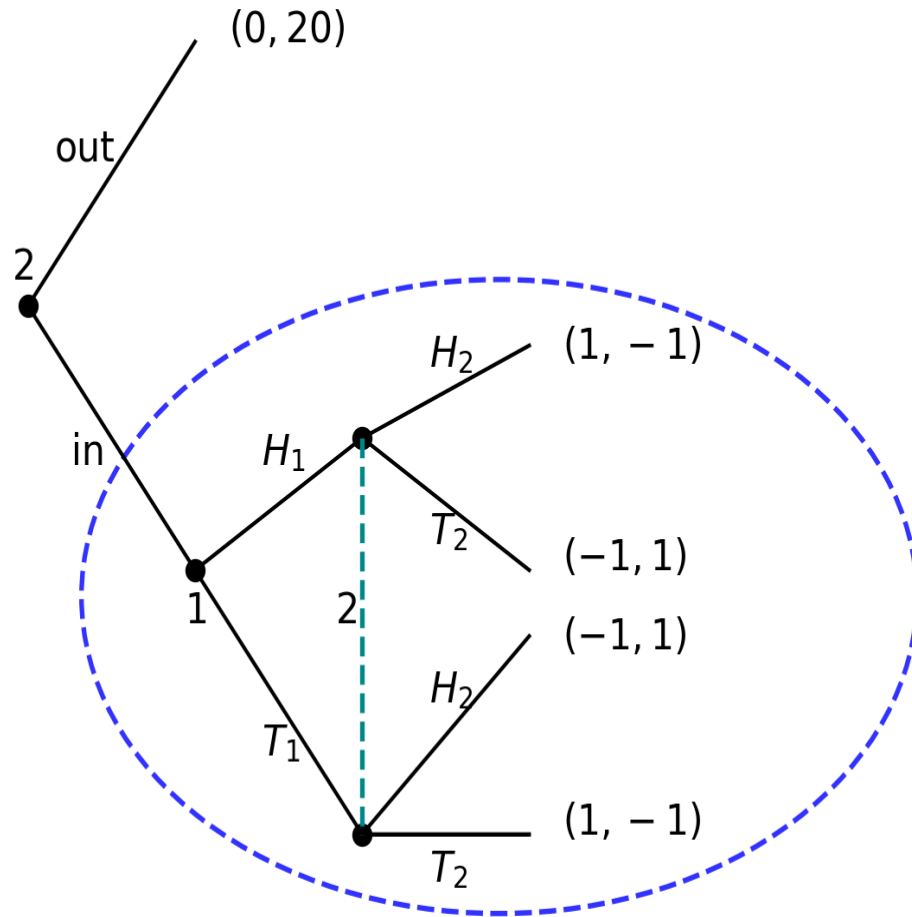
See Maschler, Solan, and Zamir (2013), page 254.

Definition 1 (Equilibrium path) Given Γ and a behavior strategy profile σ , the *path of play* according to σ is the set of nodes that are reached with strictly positive probability when players move according to σ .

If σ is an equilibrium, the path of play is termed the *equilibrium path*.

Path of play and equilibrium path

Matching pennies with outside option



Matching pennies. NE: $\sigma_1 = (1/2, 1/2) = \sigma_2$

-
- Given Γ , $\mathcal{U}_i$ is the set of player i 's information sets in Γ .

Definition 2 An information set $h_i \in \mathcal{U}_i$ is said to be *in the path of play* for a profile σ , if it is reached with positive probability according to σ , i.e.
 $P_\sigma(h_i) > 0$.

NE and Subgame Perfect equilibrium

Theorem 1 If a strategy profile σ is a NE of an extensive form game Γ , then σ induces a NE in any subgame $\Gamma(x)$ where x is in the equilibrium path of Γ (that is to say $P_\sigma(x) > 0$).

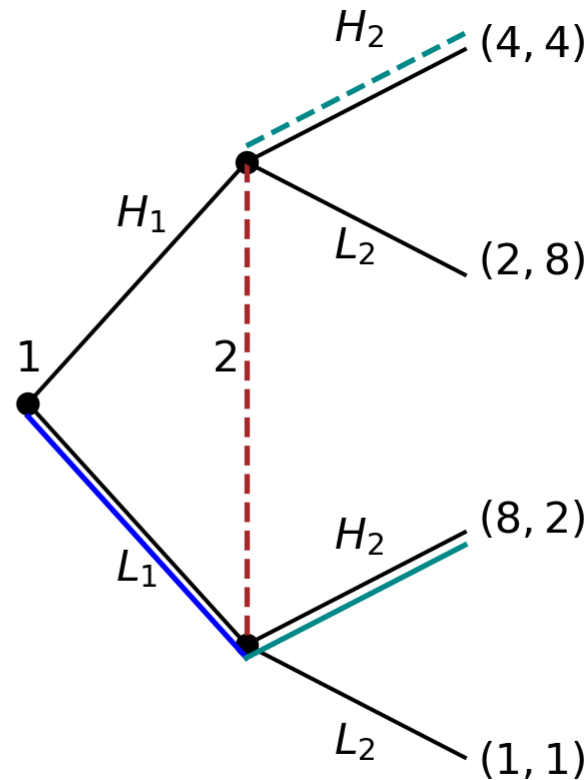
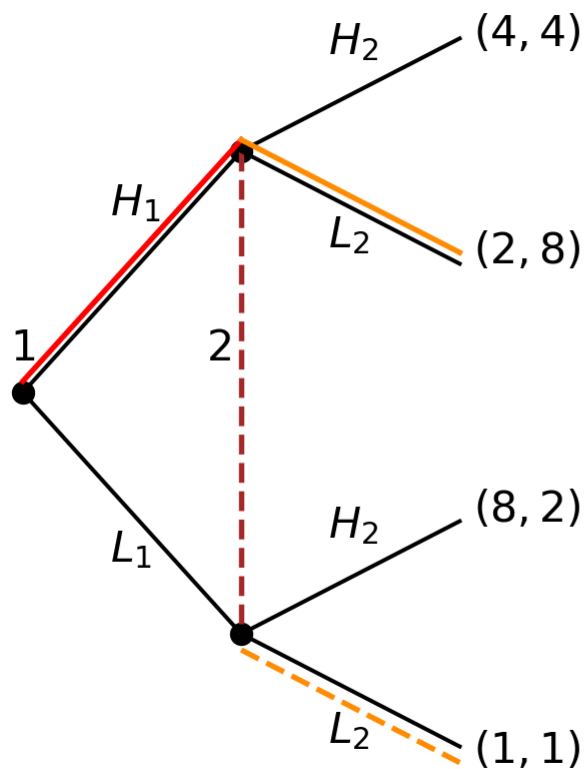
For instance, Maschler, Solan, and Zamir (2013), Theorem 7.5, page 254.

Definition 3 (Subgame perfect) A strategy profile σ is a Subgame Perfect Equilibrium (SPE) of an extensive form game Γ if σ induces a NE in every subgame of Γ .

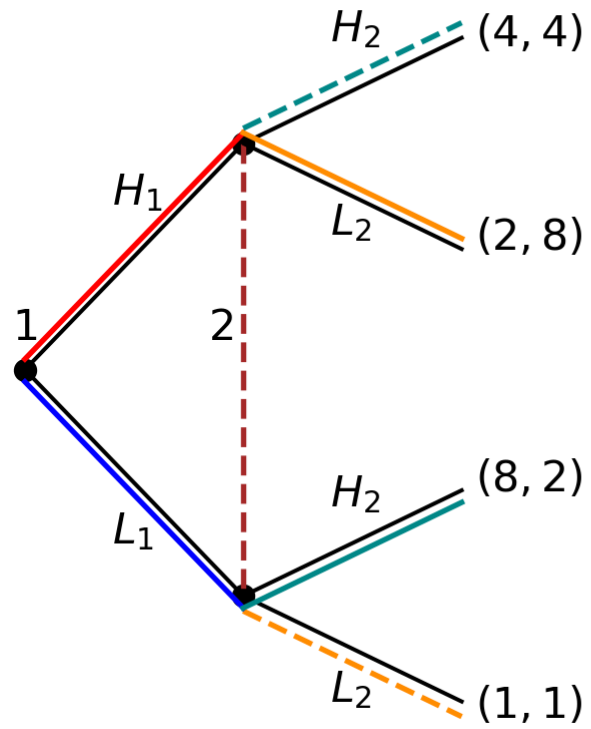
Theorem 2 Every SPE of Γ is also a NE.

Sequential games and subgame perfection

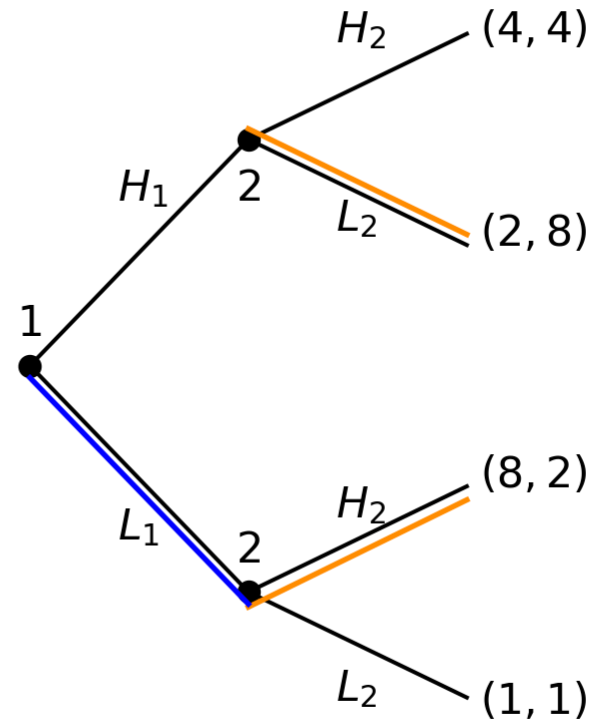
	L_2	H_2
L_1	4, 4	2, 8
H_1	8, 2	1, 1



Each player i has a single information set; thus $S_i = \{L_i, H_i\}$.

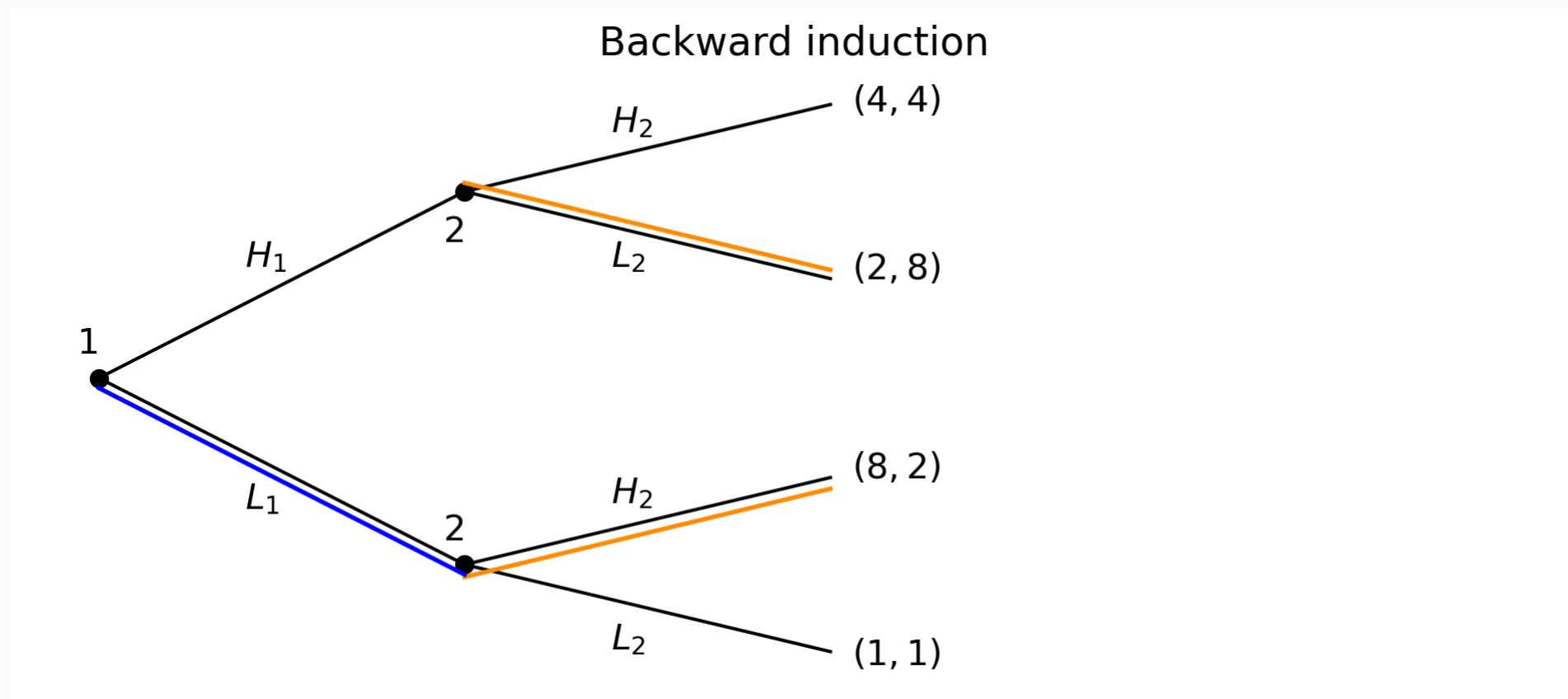


Backward induction



Subgame perfection–Example

	(H_2, H_2)	(H_2, L_2)	(L_2, H_2)	(L_2, L_2)
H_1	4,4	4,4	2,8	[2,8]
L_1	[8,2]	1,1	[8, 2]	1,1



Strategy player 2: (H_2, L_2) indicates H_2 if one played H_1 .

Completely mixed

Definition 4 (Completely mixed)

- A mixed strategy μ_i for player i in a finite normal form game is completely mixed if $\mu_i(s) > 0$ for every pure strategy s of player i .
- A strategy profile μ is completely mixed if μ_i is completely mixed for every i .
- A behavioral strategy σ_i for player i in a finite extensive form game is completely mixed if for every information set $h \in \mathcal{U}_i$, and every action $a \in C(h)$, $\sigma_i(a|h) > 0$.
- A strategy profile σ is completely mixed if σ_i is completely mixed for every i .

Completely mixed NE and SPE

Theorem 3 Every NE in completely mixed strategies of a finite extensive-form game is a SPE.

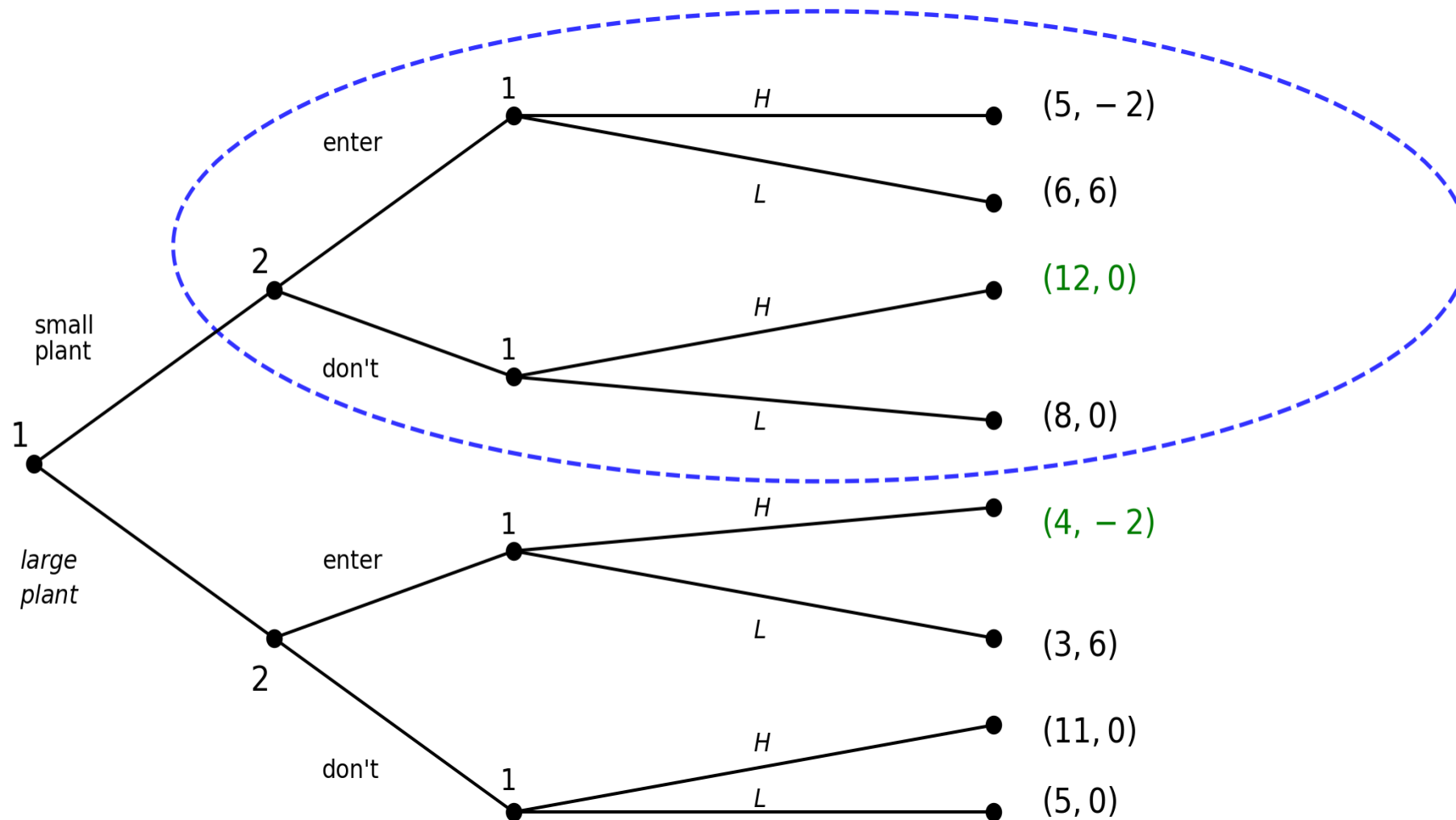
See Maschler, Solan, and Zamir (2013), Corollary 7.7.

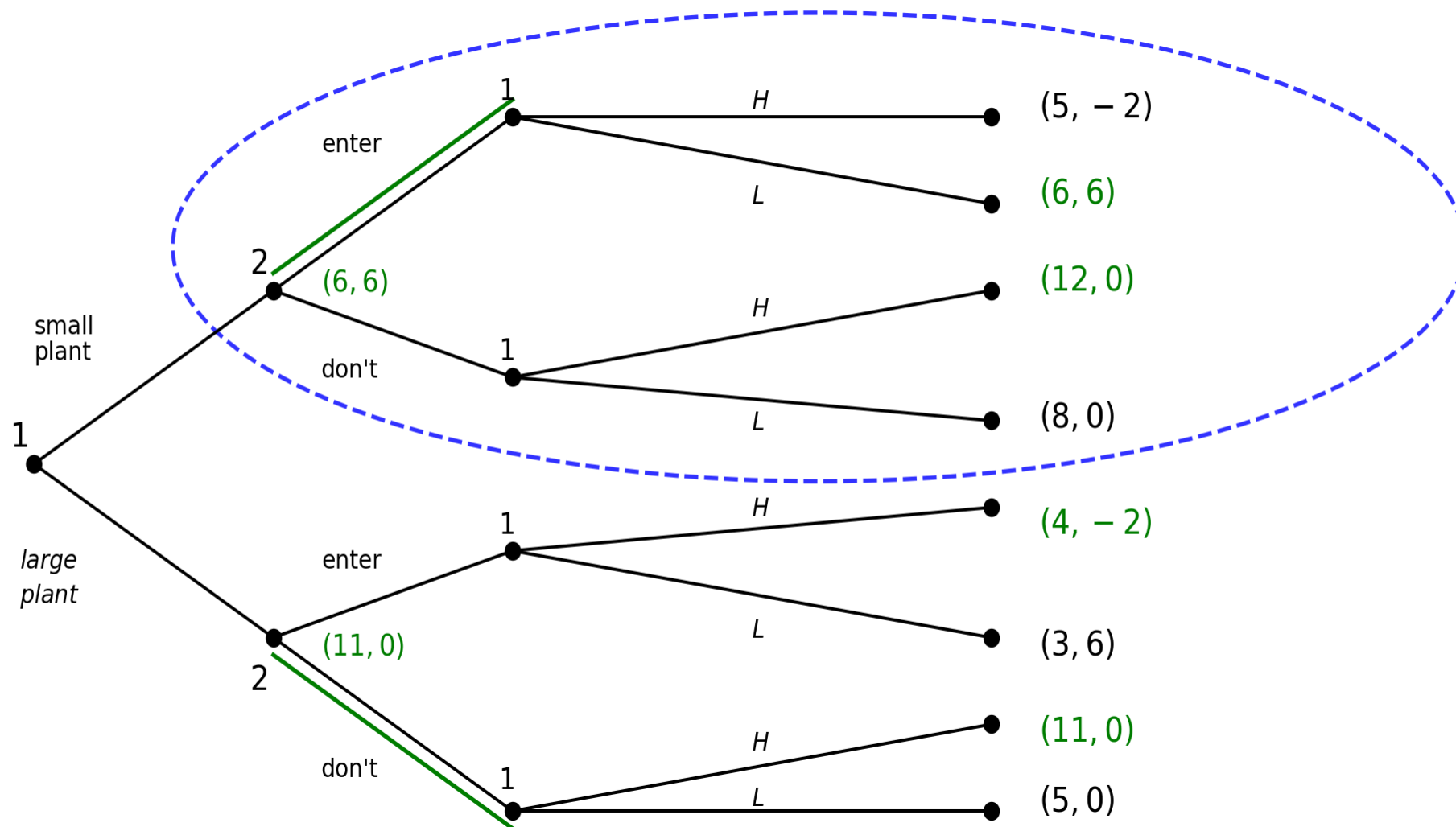
Theorem 4 Every finite extensive-form game with perfect information has a SPE in pure strategies.

Proof is a backward induction argument.

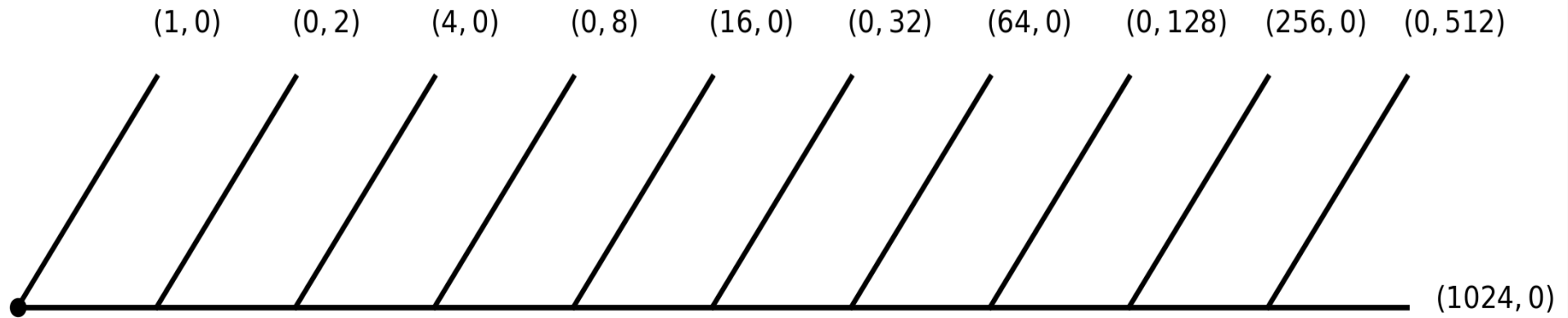
See Maschler, Solan, and Zamir (2013), Theorem 7.9.

Example: Entry deterrence





Grab the Dollar



References

Maschler, Michael, Eilon Solan, and Shmuel Zamir. 2013. *Game Theory*. Translated by Ziv Hellman. 1st ed. Cambridge University Press.